

화학수학

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확률

Probability Distribution (확률분포)

mathematical function that gives
the probabilities of occurrence of
different possible outcomes for an experiment

- Discrete probability distribution (이산 확률분포):
possible outcomes have discrete values
- Continuous probability distribution (연속 확률분포):
possible outcomes have continuous values

Probability Distributions

	Discrete	Continuous
Variable	$x_1, x_2, \dots, x_n$	x : real in $[x_{\min}, x_{\max}]$
Probability distribution	p_i	$f(x) dx$
Normalization	$\sum_{i=1}^n p_i = 1$	$\int_{x_{\min}}^{x_{\max}} f(x) dx = 1$
Mean value (normalized dist.)	$\langle x \rangle = \sum_{i=1}^n x_i p_i$	$\langle x \rangle = \int_{x_{\min}}^{x_{\max}} x f(x) dx$
Expected value (normalized dist.)	$\langle g \rangle = \sum_{i=1}^n g(x_i) p_i$	$\langle g \rangle = \int_{x_{\min}}^{x_{\max}} g(x) f(x) dx$

Examples

- Discrete probability distribution
 - Binomial probability distribution
- Continuous probability distribution
 - Uniform probability distribution
 - Gaussian probability distribution

Binomial Probability Distribution

- p = probability that “heads” will occur
- $(1 - p)$ = probability that “tails” will occur

$$P(n, m) = \frac{n!}{m! (n - m)!} p^m (1 - p)^{n-m}$$

Mean Value and Variance

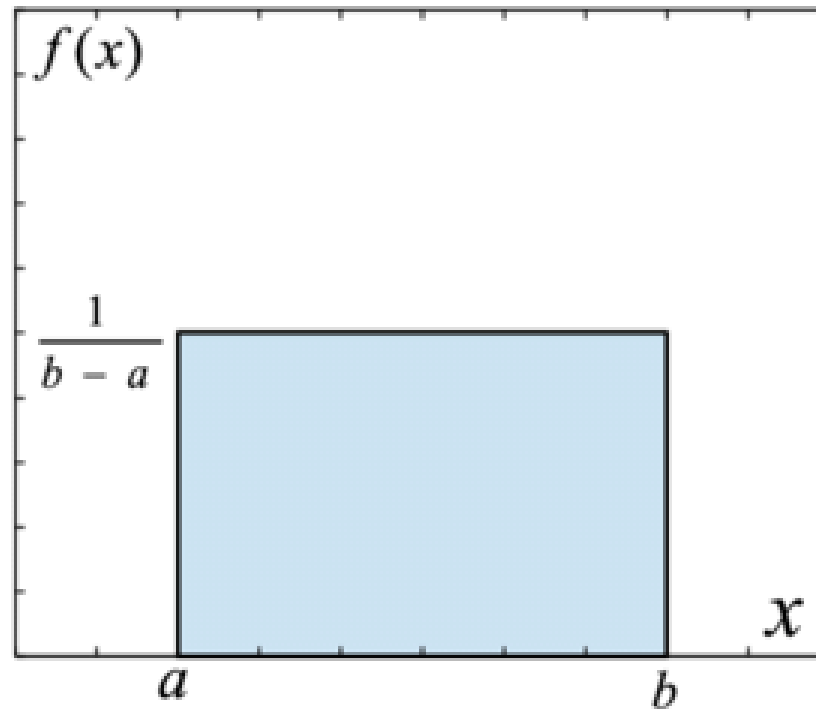
- Mean value

$$\langle x \rangle = \sum_{i=1}^{\infty} p_i x_i = \sum_{m=0}^n \frac{n!}{m! (n-m)!} p^m (1-p)^{n-m} \times m$$
$$\langle x \rangle = np$$

- Variance

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = np(1-p)$$

Uniform Probability Distribution



Mean Value and Variance

- Mean value

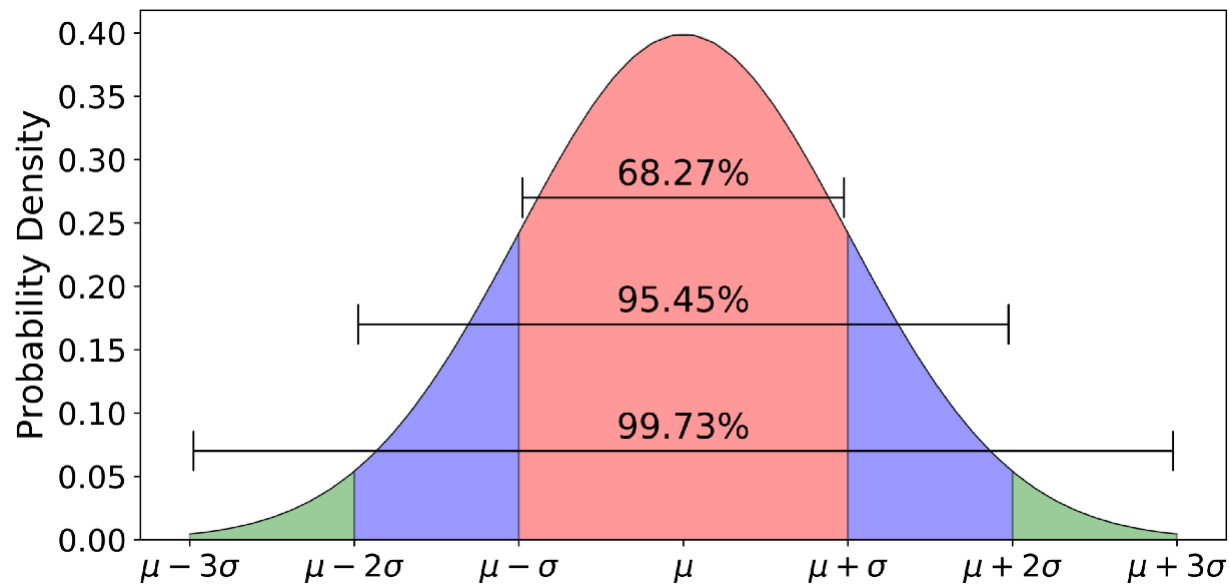
$$\langle x \rangle = \frac{a + b}{2}$$

- Variance

$$\sigma_x^2 = \frac{(a - b)^2}{12}$$

Gaussian Probability Distribution

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right]$$



(Image: <https://towardsdatascience.com/understanding-the-68-95-99-7-rule-for-a-normal-distribution-b7b7cbf760c2>)

Mean Value and Variance

- Mean value

$$\langle x \rangle = \int_{-\infty}^{\infty} x f(x) dx = \mu$$

- Variance

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \sigma^2$$

Central Limit Theorem (중심극한정리)

For random variables $x_1, x_2, \dots, x_n$,

$$y = x_1 + x_2 + \dots + x_n$$

is governed by a probability distribution that approaches
a Gaussian distribution for large n

- If experimental errors arise from multiples sources, they should be at least approximately described by a Gaussian distribution

오늘의 수업

- Applications of probability theory
 - Quantum mechanics
 - Kinetic theory of a gas

Application: Quantum Mechanics

For a wave function ψ of the system,

$$\text{Probability distribution} \propto |\psi|^2$$

e.g. Ground state of a 1-dimensional particle in a box

- Wave function $\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$
- Probability distribution $f(x) = |\psi|^2 = \frac{2}{L} \sin^2\left(\frac{\pi x}{L}\right)$

Modified Example 15.9

For a particle in a box of length L in its ground state, find the probability that the particle will be found in $(0, L/2)$.

$$\text{Probability} = \int_0^{L/2} f(x) \, dx = \int_0^{L/2} |\psi(x)|^2 \, dx$$

$$\text{Since } f(x) = |\psi|^2 = \frac{2}{L} \sin^2 \left(\frac{\pi x}{L} \right),$$

$$\text{Probability} = \int_0^{L/2} \frac{2}{L} \sin^2 \left(\frac{\pi x}{L} \right) \, dx = \frac{2}{\pi} \int_0^{\pi/2} \sin^2 t \, dt = \frac{1}{2}$$

Operators in Quantum Mechanics

“For every **observable quantity**,
there is a corresponding **mathematical operator**”

- Position operators: $\hat{x} = x$, $\hat{y} = y$, $\hat{z} = z$
- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \quad \hat{p}_y = \frac{\hbar}{i} \frac{\partial}{\partial y}, \quad \hat{p}_z = \frac{\hbar}{i} \frac{\partial}{\partial z}$$

- System energy operator (Hamiltonian):

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(x, y, z)$$

Quantum Measurement

If a wavefunction ψ of the system satisfies

$$\hat{A}\psi = a\psi$$

for a certain operator $\hat{A}$,

the **measurement** of the corresponding observable always gives a (eigenvalue).

If Wavefunction $\neq$ Eigenfunction?

If a wavefunction is represented as **linear combination** of the eigenfunctions of an operator ($\hat{A}f_k = a_k f_k$),

$$\psi = c_1 f_1 + c_2 f_2 + c_3 f_3 + \cdots + c_n f_n,$$

the measurement gives the eigenvalue a_i with the **probability** of the squared coefficient $|c_i|^2$.

Expectation Value (기댓값)

“The predicted mean value of
many measurements of an observable”
(expected value)

For $\psi = c_1 f_1 + c_2 f_2 + c_3 f_3 + \cdots + c_n f_n$,

$$\begin{aligned}\langle A \rangle &= a_1 P_1 + a_2 P_2 + \cdots + a_n P_n \\ &= a_1 |c_1|^2 + a_2 |c_2|^2 + \cdots + a_n |c_n|^2\end{aligned}$$

Expectation Value (기댓값)

For a wave function ψ of the system,

$$\langle A \rangle = \int \psi^* \hat{A} \psi \, dq$$

where $\hat{A}$ is the operator corresponding to the variable A .

Example 15.8

Find the expressions for $\langle x \rangle$ and σ_x for a particle in a box of length L in its ground state.

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$$

The expectation values are

$$\langle x \rangle = \int_0^L \psi^*(x) \hat{x} \psi(x) dx = \int_0^L x |\psi|^2 dx = \frac{2}{L} \int_0^L x \sin^2\left(\frac{\pi x}{L}\right) dx = \frac{L}{2}$$

$$\langle x^2 \rangle = \int_0^L \psi^*(x) \hat{x}^2 \psi(x) dx = \int_0^L x^2 |\psi|^2 dx = \frac{2}{L} \int_0^L x^2 \sin^2\left(\frac{\pi x}{L}\right) dx = \frac{L^2}{3} - \frac{L^2}{2\pi^2}$$

$$\sigma_x = \sqrt{\sigma_x^2} = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{L^2}{3} - \frac{L^2}{2\pi^2} - \frac{L^2}{4}} = L \sqrt{\frac{1}{12} - \frac{1}{2\pi^2}}$$

1-Dimensional Hamiltonian Operator

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \mathcal{V}(x)$$

- Corresponds to the system energy
- For a particle in a box, $\mathcal{V}(x) = 0$

Example 15.10

Find the expectation value of the energy and its standard deviation for the ground state of our particle in a box.

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right); \quad \hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$$

$$\begin{aligned} \langle E \rangle &= \int_0^L \psi^*(x) \hat{H} \psi(x) dx = \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \left\{ \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \right\} \right] dx \\ &= -\frac{\hbar^2}{mL} \int_0^L \sin\left(\frac{\pi x}{L}\right) \left(-\frac{\pi^2}{L^2} \right) \sin\left(\frac{\pi x}{L}\right) dx = \frac{\pi^2 \hbar^2}{mL^3} \int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx \\ &= \frac{\pi^2 \hbar^2}{mL^3} \times \frac{L}{2} = \frac{\pi^2 \hbar^2}{2mL^2} = \frac{h^2}{8mL^2} \end{aligned}$$

Example 15.10

Find the expectation value of the energy and its standard deviation for the ground state of our particle in a box.

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right); \quad \hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$$

$$\begin{aligned} \langle E^2 \rangle &= \int_0^L \psi^*(x) \hat{H}^2 \psi(x) dx = \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \left[\left(\frac{\hbar^2}{2m}\right)^2 \frac{d^4}{dx^4} \left\{ \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \right\} \right] dx \\ &= \frac{\hbar^4}{2m^2 L} \int_0^L \sin\left(\frac{\pi x}{L}\right) \left(\frac{\pi^4}{L^4}\right) \sin\left(\frac{\pi x}{L}\right) dx = \frac{\pi^4 \hbar^4}{2m^2 L^5} \int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx \\ &= \frac{\pi^4 \hbar^4}{2m^2 L^5} \times \frac{L}{2} = \frac{\pi^4 \hbar^4}{4m^2 L^4} = \frac{h^4}{64m^2 L^4} \end{aligned}$$

Example 15.10

Find the expectation value of the energy and its standard deviation for the ground state of our particle in a box.

$$\langle E \rangle = \frac{h^2}{8mL^2}; \quad \langle E^2 \rangle = \frac{h^4}{64m^2L^4}$$

The standard deviation is

$$\sigma_E = \sqrt{\sigma_E^2} = \sqrt{\langle E^2 \rangle - \langle E \rangle^2} = \sqrt{\frac{h^4}{64m^2L^4} - \left(\frac{h^2}{8mL^2}\right)^2} = 0$$

Application: Kinetic Theory of a Gas

- **Boltzmann factor:** in the thermal equilibrium of temperature T , the probability of having energy E is

$$f(E) \propto e^{-E/k_B T}$$

- The probability that a gas molecule has the velocity (v_x, v_y, v_z) is

$$f(v_x, v_y, v_z) \propto \exp\left(-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right)$$

Normalization

$$f(v_x, v_y, v_z) = A \exp\left(-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right)$$

$$I = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(v_x, v_y, v_z) dv_x dv_y dv_z = 1$$

$$I = A \int_{-\infty}^{\infty} e^{-mv_x^2/2k_B T} dv_x \int_{-\infty}^{\infty} e^{-mv_y^2/2k_B T} dv_y \int_{-\infty}^{\infty} e^{-mv_z^2/2k_B T} dv_z$$

$$I = A \left(\sqrt{\frac{2\pi k_B T}{m}} \right)^3 \rightarrow A = \left(\frac{m}{2\pi k_B T} \right)^{3/2}$$

Velocity to Speed

- $f(v_x, v_y, v_z) dv_x dv_y dv_z$:

Probability that a molecule has a velocity in the range

$$[v_x, v_x + dv_x], [v_y, v_y + dv_y], [v_z, v_z + dv_z]$$

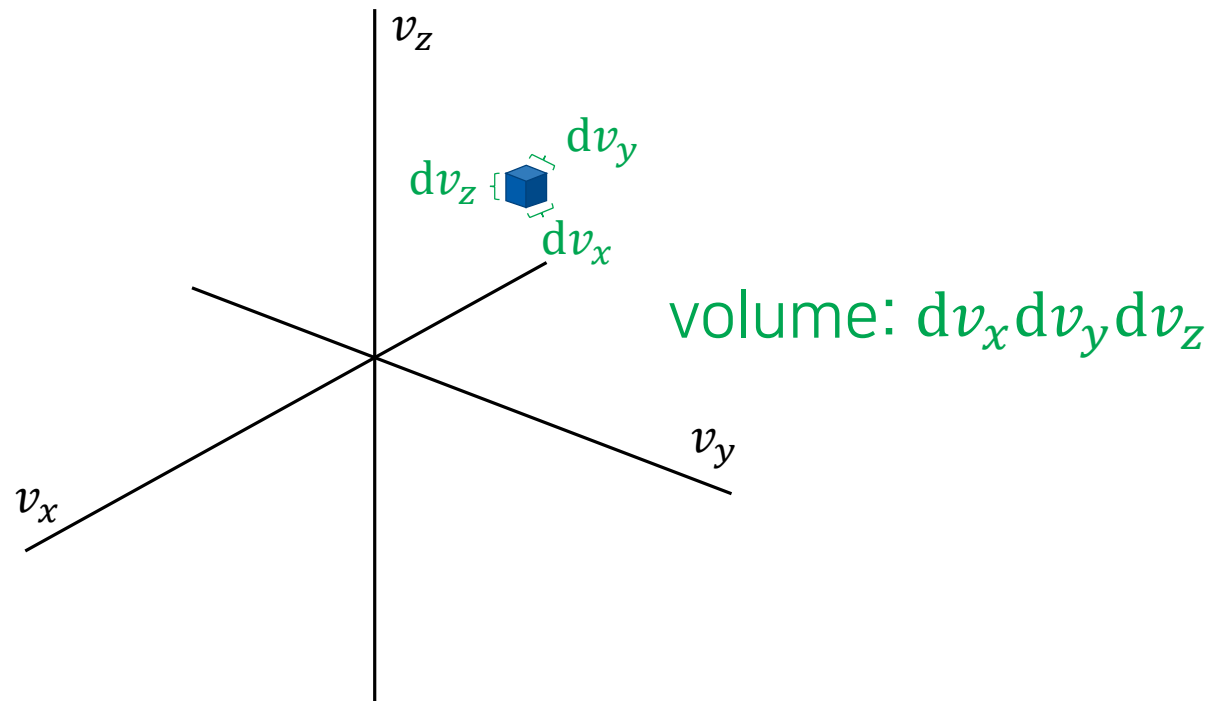
- $f_v(v) dv$:

Probability that a molecule has a speed in the range

$$[v, v + dv]$$

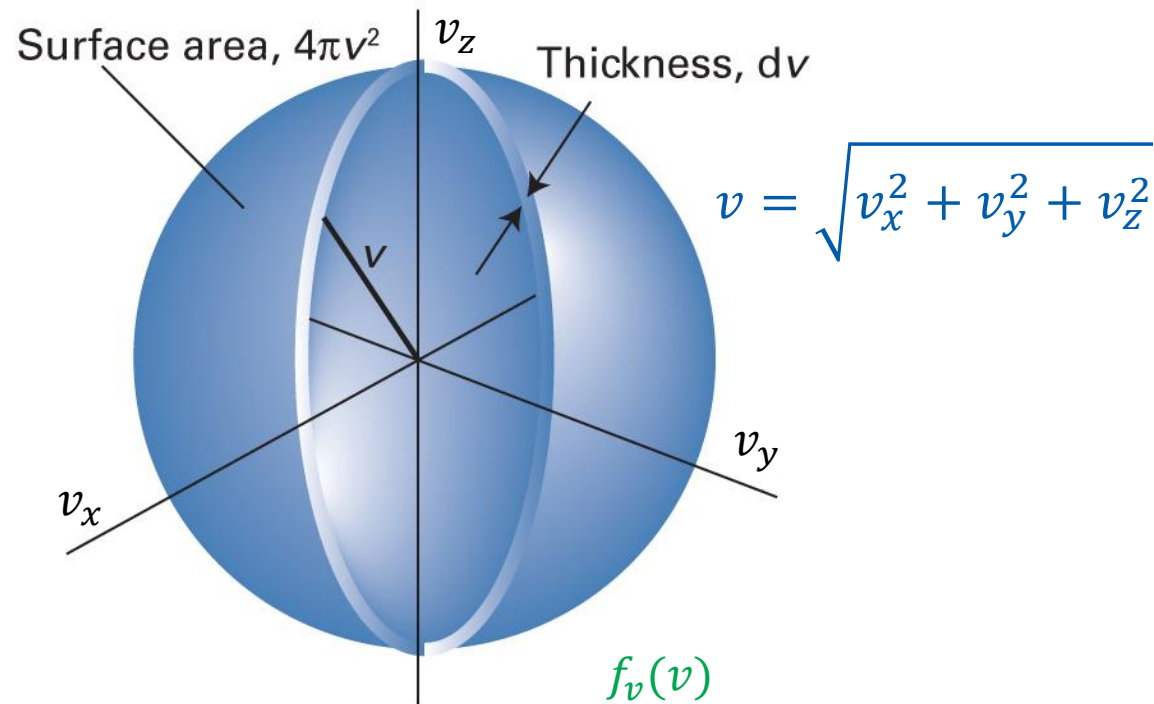
Can we obtain $f_v(v)dv$ from $f(v_x, v_y, v_z)dv_x dv_y dv_z$?

Velocity Space



$$f(v_x, v_y, v_z) dv_x dv_y dv_z = \text{function} \times \text{volume}$$

Velocity Space



$$\text{function} \times \text{volume} = \overbrace{f(v_x, v_y, v_z)}^{f_v(v)} \times 4\pi v^2 dv$$

(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1B.3)

Maxwell-Boltzmann Distribution

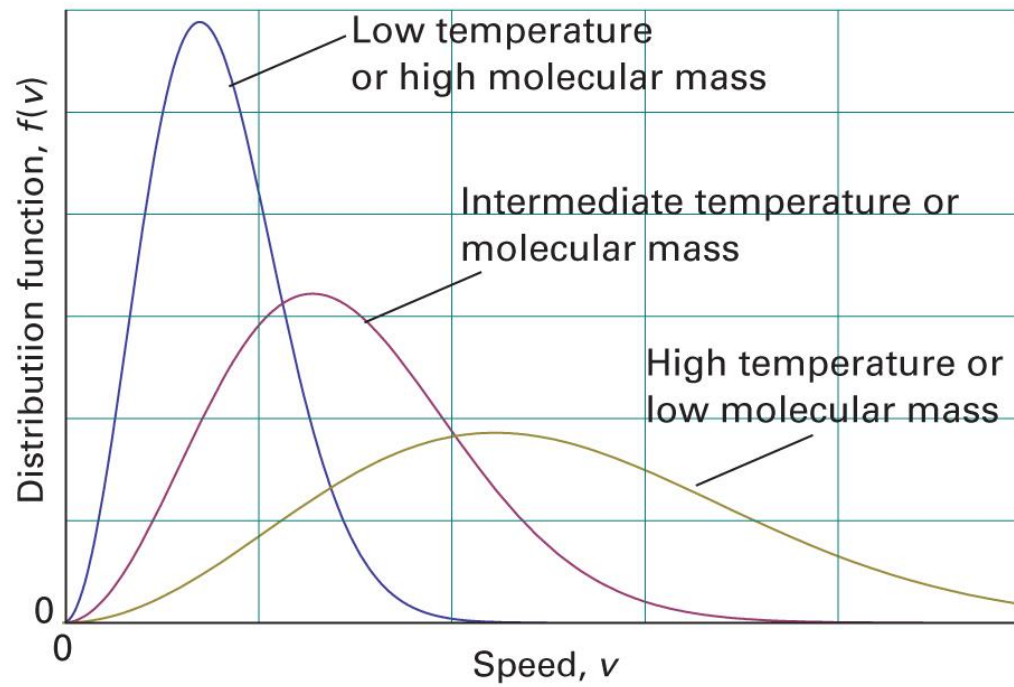
$$f_v(v) = f(v_x, v_y, v_z) 4\pi v^2$$

$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-m(v_x^2 + v_y^2 + v_z^2)/2k_B T}$$

$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$

Behaviors of the Distribution

$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1B.4)

Mean Speed (평균 속력)

$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$

$$\langle v \rangle = \int_0^\infty v f_v(v) dv = \int_0^\infty 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^3 e^{-mv^2/2k_B T} dv$$

$$= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^3 e^{-mv^2/2k_B T} dv$$

$$= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \frac{(2k_B T)^2}{2m^2} = \left(\frac{8k_B T}{\pi m} \right)^{1/2}$$

Example 15.12

Find a formula for the mean of the square of the speed and the root-mean-square speed of molecules in a gas.

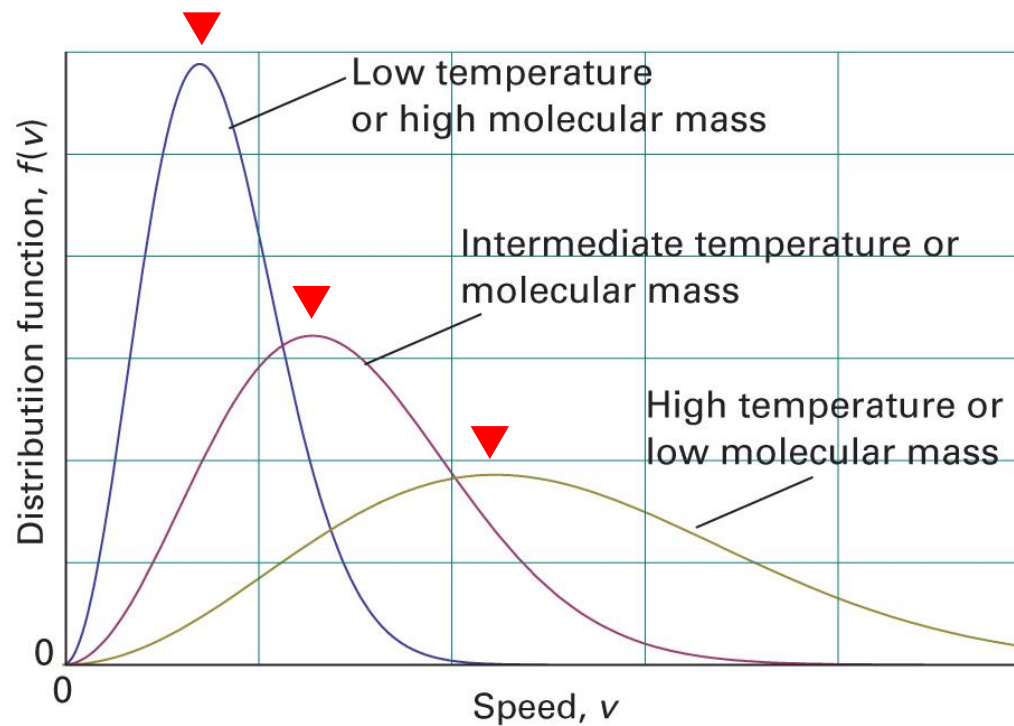
$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$

$$\langle v^2 \rangle = \int_0^\infty v^2 f_v(v) dv = \int_0^\infty 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^4 e^{-mv^2/2k_B T} dv$$

$$= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^4 e^{-mv^2/2k_B T} dv = \frac{3k_B T}{m}$$

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle} = \left(\frac{3k_B T}{m} \right)^{1/2}$$

Most Probable Speed (최빈 속력)



$$\frac{df(v)}{dv} = 0$$

(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1B.4)

Most Probable Speed (최빈 속력)

$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$

$$\frac{df_v(v)}{dv} = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \left(2v e^{-mv^2/2k_B T} - \frac{mv^3}{k_B T} e^{-mv^2/2k_B T} \right) = 0$$

$$2v - \frac{mv^3}{k_B T} = 0$$

$$v_{\text{mp}} = \left(\frac{2k_B T}{m} \right)^{1/2}$$